



SOLUTIONS OF INTRODUCTION TO CHEMISTRY

EXERCISE

PART - I

- 10^{10} grains are distributed in 1 second

$\therefore 6.02 \times 10^{23}$ grains are distributed in $\frac{6.02 \times 10^{23}}{10^{10}} \text{ sec} = \frac{6.02 \times 10^{23}}{10^{10} \times 60 \times 60 \times 24 \times 365} \text{ years}$

$= 1.9 \times 10^6 \text{ years (approx.)}$
- No. of atoms = mole $\times N_a$

$1 = \frac{x}{238} \times N_a$ (x is wt. of uranium)

$x = \frac{238}{6} \times 10^{-23}$

$x = 3.95 \times 10^{-22}$
- No. of moles of C = $\frac{12.044 \times 10^{23}}{6.022 \times 10^{23}} = 2$.

Wt. of C atoms = $2 \times 12 = 24 \text{ g}$.
- mass of Si = mole $\times$ Atomic mass

$= 35 \times 28 = 980 \text{ g}$
- We know that, $1 \text{ amu} = \frac{1}{12} \times \text{weight of one } ^{12}\text{C atom}$

or weight of one $^{12}\text{C atom} = 12 \text{ amu}$ (at. wt. of C = 12 amu).

Similarly, as the atomic weight of He is 4 amu,

weight of one He atom = 4 amu.

Thus, the number of atoms in 100 amu of He = $\frac{100}{4} = 25$.
- 1 litre Hg metal

volume = 1000

$d = \frac{m}{v}$ mass = $d \times V = 13.6 \times 1000$

No of mole of Hg metal = $\frac{13.6 \times 1000}{200} = 68 \text{ mole}$
- Fractional abundance of $^{35}\text{Cl} = 0.75$, Molar mass = 35.0

Fractional abundance of $^{37}\text{Cl} = 0.25$, Molar mass = 37.0

$\therefore$ Average atomic mass = $(0.75) (35.0 \text{ amu}) + (0.25) (37.0 \text{ amu}) = 35.5$
- Let mole % of ^{26}Mg be x.

$\therefore \frac{(21-x)25 + x(26) + 79(24)}{100} = 24.31$

$x = 10\%$
- No. of molecules = mole $\times N_a = \frac{16}{16} \times N_a$

$N_a = 6.02 \times 10^{23}$



13. In 18 g, no. of molecules = N_A
 so in 0.09 g no. of molecules = $\frac{N_A}{18} \times 0.09 = \frac{N_A}{2 \times 100} = 3.01 \times 10^{21}$.
14. Let the number of C_2H_6 molecules in the sample be n . As given, mass of C_2H_6 = mass of 10^7 molecules of CH_4

$$\frac{n}{\text{Av. constant}} \times \text{mol. wt. of } C_2H_6 = \frac{10^7}{\text{Av. constant}} \times \text{mol. wt. of } CH_4$$

$$\frac{n \times 30}{\text{Av. constant}} = \frac{10^7 \times 16}{\text{Av. constant}} = 5.34 \times 10^6.$$
16. No. of moles of $CaCO_3 = \frac{\text{no. of molecules}}{\text{Av. cons.}} = \frac{6.022 \times 10^{23}}{6.022 \times 10^{23}} = 1$
 Weight of $CaCO_3 = 1 \times 100 = 100$ g.
17. Total no. of moles of $CO_2 = \frac{\text{wt. in g}}{\text{mol. wt.}} = \frac{0.2}{44} = 0.00454$.
 No. of moles removed = $\frac{10^{21}}{6.022 \times 10^{23}} = 0.00166$.
 No. of moles of CO_2 left = $0.00454 - 0.00166 = 0.00288$.
18. (a) mole of $H_2SO_4 = \frac{\text{mass}}{\text{molar mass}} = \frac{196}{98} = 2$.
 1 molecule H_2SO_4 contains 2 atom hydrogen, 1 atom sulphur and 4 atom of oxygen.
 Hence, H = $4N_A$ atoms, S = $2N_A$ atoms, O = $8N_A$ atoms
 (b) molecule of $H_2SO_4 = \frac{196}{98} = 2$.
 Hence, H = 4 atoms, S = 2 atoms, O = 8 atoms.
 (c) 5 mole $H_2S_2O_8$ contains
 H = $10N_A$ atoms, S = $10N_A$ atoms, O = $40N_A$ atoms
 (d) 3 molecules $H_2S_2O_6$ contains
 H = 6 atoms, S = 6 atoms, O = 18 atoms.
19. 10 mole NH_3 have mole of 'H' atom = 10×3
 5 mole of H_2SO_4 have mole of 'H' atom = 10
 Total mole of 'H' atom = 40
 mole of $H_2 = 20$
 Hence: number of H_2 molecules = $20N_A$
20. no. of atoms = $3 \times 11 \times N_A$
 So no. of O_3 molecules formed = $11N_A$
21. Mol. wt. of air = $\frac{78 \times 28 + 21 \times 32 + 0.9 \times 40 + 0.1 \times 44}{78 + 21 + 0.9 + 0.1} = 28.964$.
 ($N_2 = 28$, $O_2 = 32$, Ar = 40 and $CO_2 = 44$)
22. From ideal gas equation, $pV = nRT$. In SI sytem the parameters of the gas are:
 Pressure = $p \times 1000$ (Pa); Volume = $V \times 10^{-6}$ (m^3); Temperature = $t + 273$ (K); moles = $w/32$
 Therefore, $R = \frac{32pV}{1000 \times w \times (t + 273)}$
23. $PV = nRT$, $N = n \times N_A$



24. $PV = nRT$,
 $n = W/M$ 16 AMU

25. Pressure = 7.6×10^{-10} mm
 $= 0.76 \times 10^{-10}$ cm
 $\frac{0.76 \times 10^{-10}}{76} = \text{atm}$ (1 atm = 76 cm) = 10^{-12} atm.

Volume = 1 litre, $R = 0.0821$ lit. atm/K/mole, temperature = 273 K.

We know that $pV = nRT$ or $n = \frac{pV}{RT}$

$$n = \frac{10^{-12} \times 1}{0.082 \times 273} = 0.44 \times 10^{-13}.$$

No. of molecules = $0.44 \times 10^{-13} \times 6.022 \times 10^{23} = 2.65 \times 10^{10}$.

PART - II

1. Atoms of an element are alike.

4. $\text{mole} = \frac{\text{mass}}{\text{at. wt.}} = \frac{46}{23} = 2 \text{ mole.}$

6. 4 g He = N_A atoms

	A	B
Atomic mass	40	80
given weight	x gram	2x gram
No. of mole	$\frac{x}{40}$	$\frac{2x}{80}$
No. of Atom	$\frac{x}{40} \times N_A$	$\frac{x}{40} \times N_A$

But according to question = $\frac{x}{40} \times N_A = y$

8. Mole of Aluminium = $\frac{54}{27} = 2 \text{ mole.}$

Al and Mg have same number of atoms (given). Hence same moles also.

$\therefore$ Mass of magnesium = $2 \times 24 = 48 \text{ g.}$

9. $558.5 \text{ g Fe} = \frac{558.5}{55.85} \text{ mole Fe} = 10 \text{ mole Fe} = 2 \times 5 \text{ mole C} = 2 \times \frac{60}{12} \text{ mole C}$

11. 12 g ${}^6\text{C}^{12}$ contains $6N_A$ electrons and $6N_A$ neutrons.

12. $M_X = 2 \times 12 = 24$

$M_Y = \frac{M_X}{0.3} = 80.$

13. 1 gram ion = 1 mole
charge on 1 mole Al^{3+} is $= 3 \times e (N_A).$

14. Number of protons in ${}^6\text{C}^{14} = 6$;
Number of neutrons in ${}^6\text{C}^{14} = 8$;
As per given new atomic mass of
 ${}^6\text{C}^{14} = 12 + 4 = 16$



(As the mass of electron negligible as compared to neutron and proton)

$$\% \text{ increase in mass} = \frac{16-14}{14} \times 100 = 14.28$$

15. Weight of C-14 isotope in 12g sample = $\frac{2 \times 12}{100}$

$$\text{No. of isotopes} = \frac{2 \times 12 \times N}{100 \times 14} = 1.032 \times 10^{22} \text{ atom}$$

17. $114.8 = 115 \times 0.95 + M \times 0.05$
 $M = 111$

19. $17 \text{ g NH}_3 = N_A \text{ molecules}$

21. Gram mol. wt. of $\text{C}_{60}\text{H}_{22} = 742 \text{ gm}$
 i.e. wt. of 6.023×10^{23} molecules = 742
 so wt. of 1 molecules = $\frac{742}{6.023 \times 10^{23}} = 1.24 \times 10^{-21} \text{ g.}$

22. Number of electrons = $\frac{1.8 \times 10}{18} \times N_A$

23. $1 \text{ mole P}_4 = N \text{ molecules of P}_4 = 4 N \text{ atoms of P}_4$.

24. In $(\text{NH}_4)_3\text{PO}_4$
 $\frac{\text{mole of H atom}}{\text{mole of O atom}} = \frac{12}{4}$
 $\text{mole of 'O' atom} = \frac{4}{12} \text{ (mole of H atom)} = \frac{1}{3} (3.18) = 1.06$.

25. (B) $1 \text{ Torr} = 1 \text{ mm.}$

27. This is the required relation in Centigrade and Fahrenheit scales.

28. $\frac{F-32}{9} = \frac{C}{5}$
 Let temperature be t , same on two scale
 $\therefore t-32 = \frac{9t}{5} \text{ or } t = -40$

30. $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1} = 8.314 \text{ JK}^{-1} \text{ mol}^{-1} = 8.314 \times 10^7 \text{ erg K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ litre atm K}^{-1} \text{ mol}^{-1}$.

31. Follow answer 1 in SI units.

33. $P \times 44.8 = 2 \times 0.0821 \times 540. \therefore P = 1.98 \text{ atm.}$

34. Molar volume, i.e. volume when $n = 1$ from $PV = nRT$ is RT/P .

37.

	H_2	:	He	:	O_2	:	O_3
Ratio of total no. of molecules =	1	:	1	:	1	:	1
So ratio of total no. of atoms =	2	:	1	:	2	:	3

38. Statement of avogadro's hypothesis.



39. Mol. wt. of gas is = $\frac{16 \times 22.4}{5.6} = 64 \text{ g}$

$$32 + 16x = 64$$

$$x = 2$$

40. $\frac{\text{wt. of 1 litre gas at STP}}{\text{wt of 1 litre O}_2 \text{ at STP}} = \frac{\text{molar mass of gas}}{\text{molar mass of O}_2}$

$$2.22 = \frac{M}{32}$$

$$M = 71.$$

43. $9.108 \times 10^{-21} \text{ kg}$ is the wt. of $1 \text{ e}^- = \frac{1}{N_A}$ moles of e^-

So 1 kg is the wt. of $1 \text{ e}^- = \frac{1}{9.108 \times 10^{-31}} \times \frac{1}{N_A} = \frac{1}{9.108 \times 10^{-31} \times 6.023 \times 10^{23}} = \frac{10^8}{9.108 \times 6.023}$

44. 560g of Fe No. of moles = $\frac{560 \text{ g}}{56 \text{ g}} = 10 \text{ mole}$

70 g of N

14g = 1 mole atom of N

70g = 5 moles of N

20g H = 20 moles of H-atoms.

45. (A) Moles of C = $24/12 = 2$, So no. of atoms = $2N_A$
 (B) Moles of Fe = $56/56 = 1$, So no. of atoms = N_A
 (C) Moles of Al = $27/27 = 1$, So no. of atoms = N_A
 (D) Moles of Fe = $108/108 = 1$, So no. of atoms = N_A

46. The mass of one mole of a substance will remain unchanged.

47. 8 moles of O-atom are contained by 1 mole $\text{Mg}_3(\text{PO}_4)_2$.

Hence, 0.25 moles of O-atom = $\frac{1}{8} \times 0.25 = 3.125 \times 10^{-2} \text{ mole Mg}_3(\text{PO}_4)_2$.

48. $^{54}\text{Fe} \longrightarrow 5\%$
 $^{56}\text{Fe} \longrightarrow 90\%$
 $^{57}\text{Fe} \longrightarrow 5\%$

Av. atomic mass = $x_1A_1 + x_2A_2 + x_3A_3 = 54 \times 0.05 + 56 \times 0.9 + 57 \times 0.05 = 55.95$

51. Number of electron = mole of H $\times 1$ = Mole of O $\times 8$ = Mole of C $\times 6$ = Mole of N $\times 7$

52. Refer Notes.

53. Pressure is same when V/T is constant.

60. Use % by moles = $\frac{M_{\text{avg}} - M_1}{M_2 - M_1} \times 100$

% by mass = % by moles $\times \frac{M_2}{M_{\text{avg}}}$